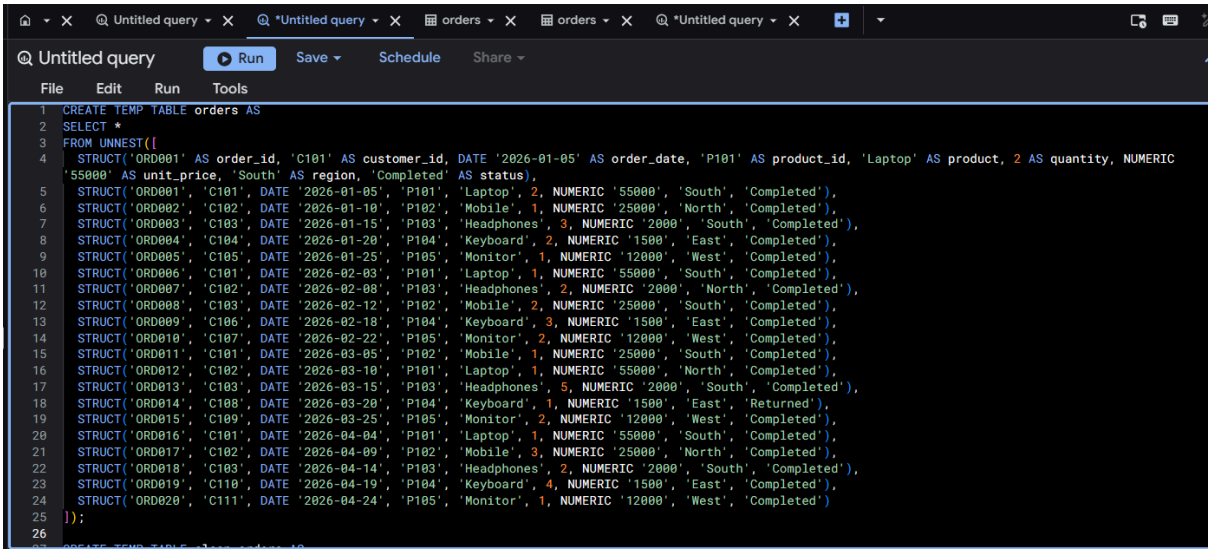
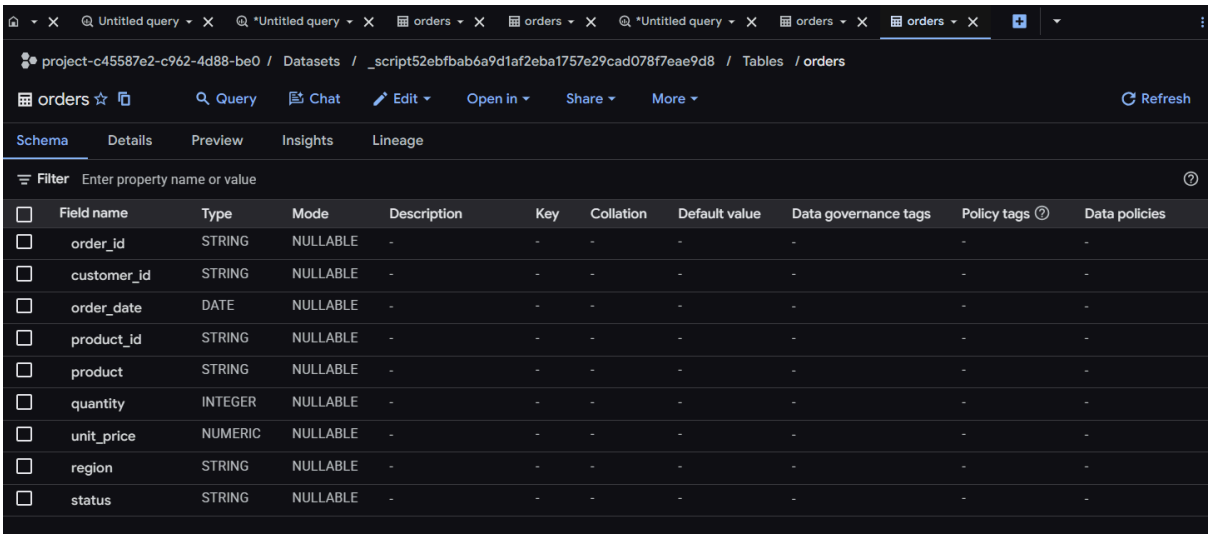
GCP - BIGQUERY

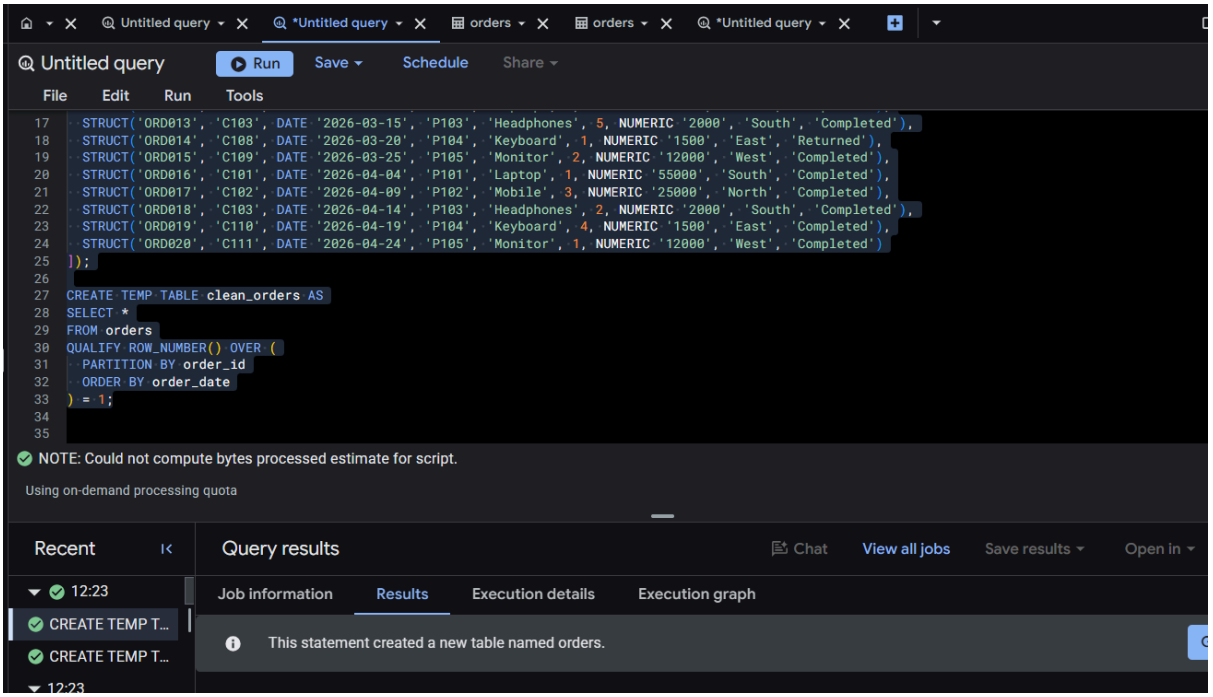
Name : M.Sainath Reddy
USN : 23BTRCL213
HCLID : HCLTFP2265832

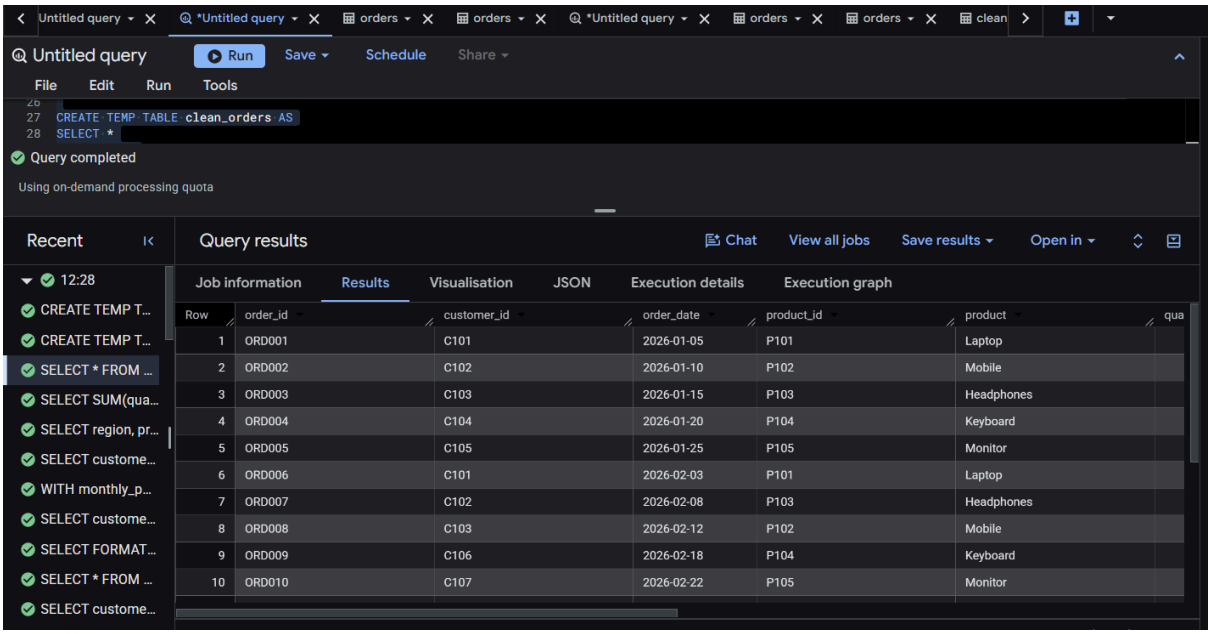
A ecommerce company receives 1000 orders daily. The business team wants to identify its highest revenue sales and top customer and monthly sale



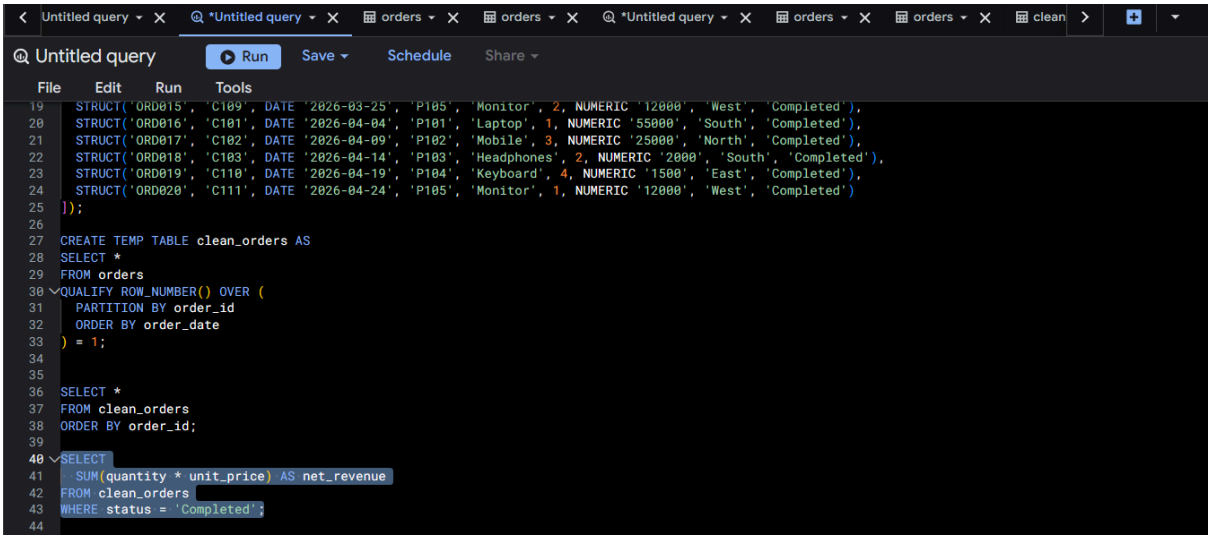


Task 1: remove the duplicate orders





Task 2: find the net revenue

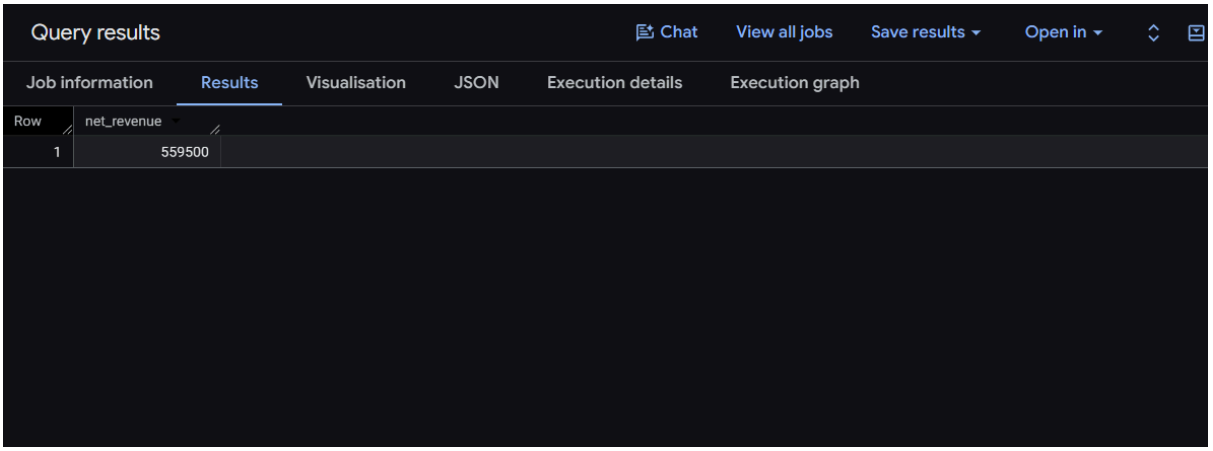


```
  STRUCT('ORD015', 'C109', DATE '2026-03-25', 'P105', 'Monitor', 2, NUMERIC '12000', 'West', 'Completed'),
  STRUCT('ORD016', 'C101', DATE '2026-04-04', 'P101', 'Laptop', 1, NUMERIC '55000', 'South', 'Completed'),
  STRUCT('ORD017', 'C102', DATE '2026-04-09', 'P102', 'Mobile', 3, NUMERIC '25000', 'North', 'Completed'),
  STRUCT('ORD018', 'C103', DATE '2026-04-14', 'P103', 'Headphones', 2, NUMERIC '2000', 'South', 'Completed'),
  STRUCT('ORD019', 'C110', DATE '2026-04-19', 'P104', 'Keyboard', 4, NUMERIC '1500', 'East', 'Completed'),
  STRUCT('ORD020', 'C111', DATE '2026-04-24', 'P105', 'Monitor', 1, NUMERIC '12000', 'West', 'Completed')
]);

CREATE TEMP TABLE clean_orders AS
SELECT *
FROM orders
QUALIFY ROW_NUMBER() OVER (
  PARTITION BY order_id
  ORDER BY order_date
) = 1;


SELECT *
FROM clean_orders
ORDER BY order_id;

SELECT
  SUM(quantity * unit_price) AS net_revenue
FROM clean_orders
WHERE status = 'Completed';
```



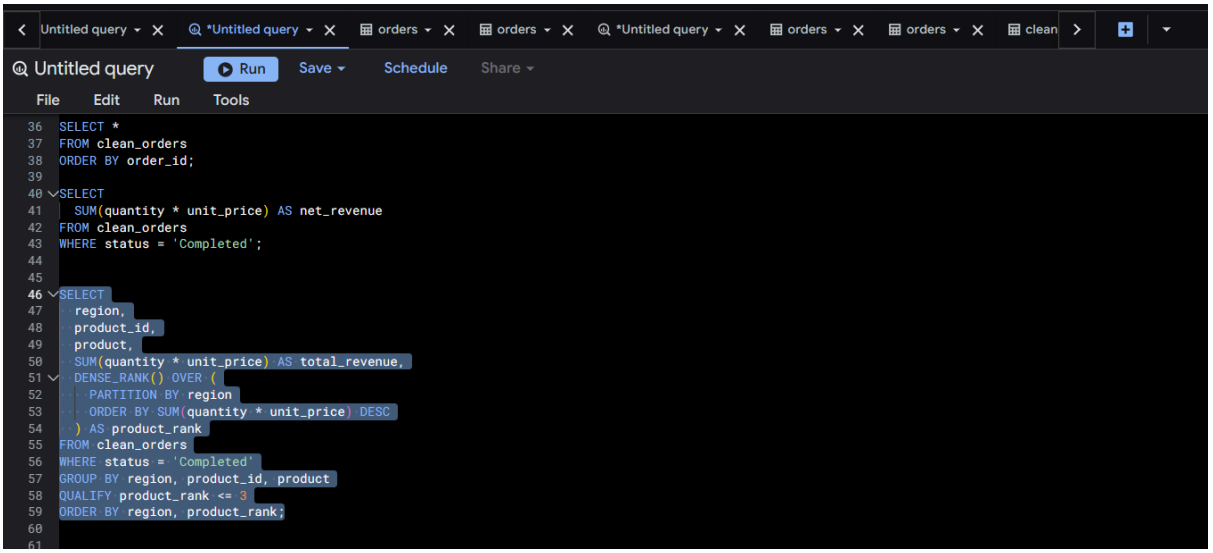
Query results

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Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	net_revenue
1	559500

Task 3: find the top 3 products in each region



```
SELECT *
FROM clean_orders
ORDER BY order_id;

SELECT
  SUM(quantity * unit_price) AS net_revenue
FROM clean_orders
WHERE status = 'Completed';


SELECT
  region,
  product_id,
  product,
  SUM(quantity * unit_price) AS total_revenue,
  DENSE_RANK() OVER (
    PARTITION BY region
    ORDER BY SUM(quantity * unit_price) DESC
  ) AS product_rank
FROM clean_orders
WHERE status = 'Completed'
GROUP BY region, product_id, product
QUALIFY product_rank <= 3
ORDER BY region, product_rank;
```

Query results

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Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	region	product_id	product	total_revenue	product_rank
1	East	P104	Keyboard	13500	1
2	North	P102	Mobile	100000	1
3	North	P101	Laptop	55000	2
4	North	P103	Headphones	4000	3
5	South	P101	Laptop	220000	1
6	South	P102	Mobile	75000	2
7	South	P103	Headphones	20000	3
8	West	P105	Monitor	72000	1

Task 4: identify inactive customers

Untitled query

Run | Save | Schedule | Share

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```
  DENSE_RANK() OVER (
    PARTITION BY region
    ORDER BY SUM(quantity * unit_price) DESC
  ) AS product_rank
FROM clean_orders
WHERE status = 'Completed'
GROUP BY region, product_id, product
QUALIFY product_rank <= 3
ORDER BY region, product_rank;



SELECT
  customer_id,
  MAX(order_date) AS last_purchase_date
FROM clean_orders
WHERE status = 'Completed'
GROUP BY customer_id
HAVING MAX(order_date) < DATE_SUB(
  CURRENT_DATE(), INTERVAL 2 MONTH
)
ORDER BY last_purchase_date;
```

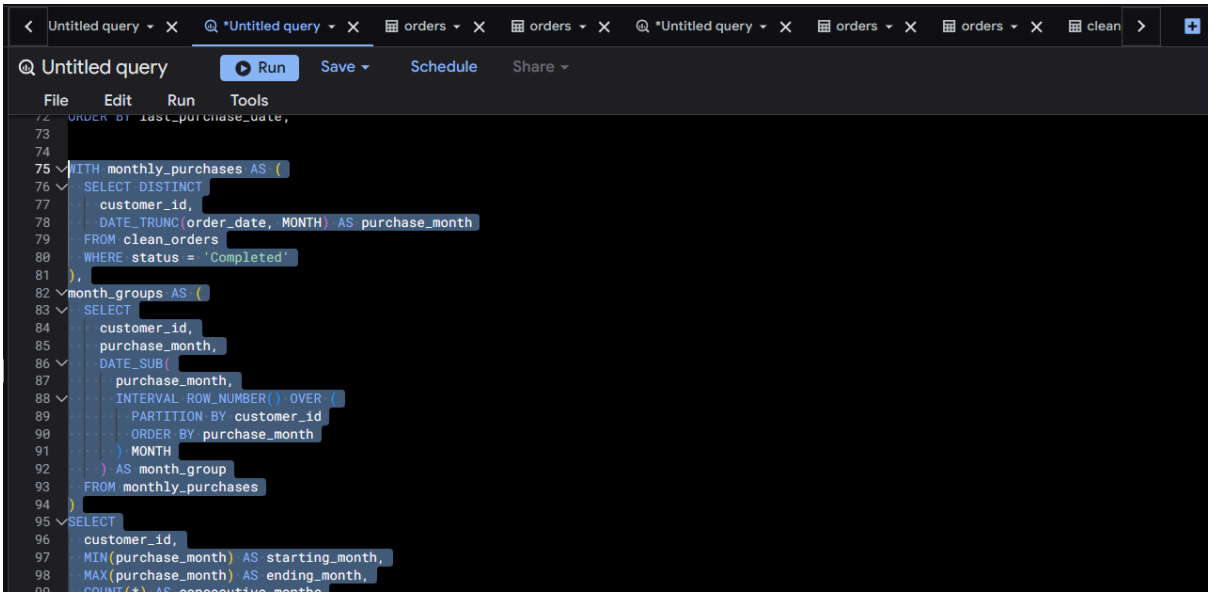
Query results

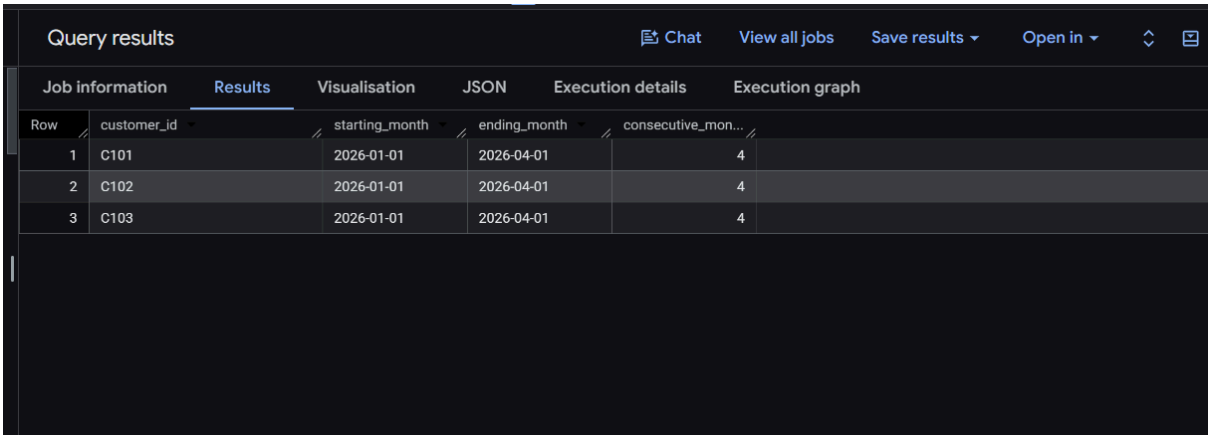
Chat | View all jobs | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_id	last_purchase_date
1	C104	2026-01-20
2	C105	2026-01-25
3	C106	2026-02-18
4	C107	2026-02-22
5	C109	2026-03-25
6	C101	2026-04-04
7	C102	2026-04-09
8	C103	2026-04-14
9	C110	2026-04-19
10	C111	2026-04-24

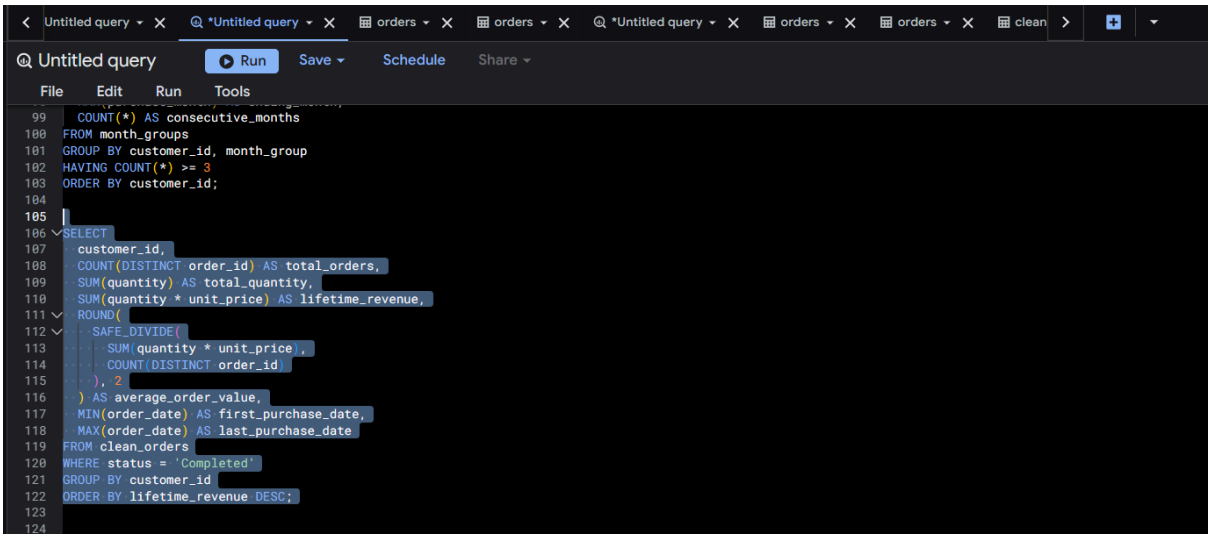
Task 5: find the customers who purchase continuously 3 months





Row	customer_id	starting_month	ending_month	consecutive_mon...
1	C101	2026-01-01	2026-04-01	4
2	C102	2026-01-01	2026-04-01	4
3	C103	2026-01-01	2026-04-01	4

Task 6: build a customer lifetime value report



Query results

Chat | View all jobs | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	customer_id	total_orders	total_quantity	lifetime_revenue	average_order_va...	first_purchase_da...	last_purchase_date
1	C101	4	5	245000	61250	2026-01-05	2026-04-04
2	C102	4	7	159000	39750	2026-01-10	2026-04-09
3	C103	4	12	70000	17500	2026-01-15	2026-04-14
4	C107	1	2	24000	24000	2026-02-22	2026-02-22
5	C109	1	2	24000	24000	2026-03-25	2026-03-25
6	C105	1	1	12000	12000	2026-01-25	2026-01-25
7	C111	1	1	12000	12000	2026-04-24	2026-04-24
8	C110	1	4	6000	6000	2026-04-19	2026-04-19
9	C106	1	3	4500	4500	2026-02-18	2026-02-18
10	C104	1	2	3000	3000	2026-01-20	2026-01-20

Task 7: find the monthly sales

Untitled query | *Untitled query | orders | orders | *Untitled query | orders | orders | clean

Untitled query — Run | Save | Schedule | Share

File | Edit | Run | Tools

```
      COUNT(DISTINCT order_id)
    ), 2
  ) AS average_order_value,
  MIN(order_date) AS first_purchase_date,
  MAX(order_date) AS last_purchase_date
FROM clean_orders
WHERE status = 'Completed'
GROUP BY customer_id
ORDER BY lifetime_revenue DESC;


SELECT
  FORMAT_DATE('%Y-%m', order_date) AS sales_month,
  COUNT(DISTINCT order_id) AS total_orders,
  SUM(quantity) AS total_quantity_sold,
  SUM(quantity * unit_price) AS monthly_revenue
FROM clean_orders
WHERE status = 'Completed'
GROUP BY sales_month
ORDER BY sales_month;
```

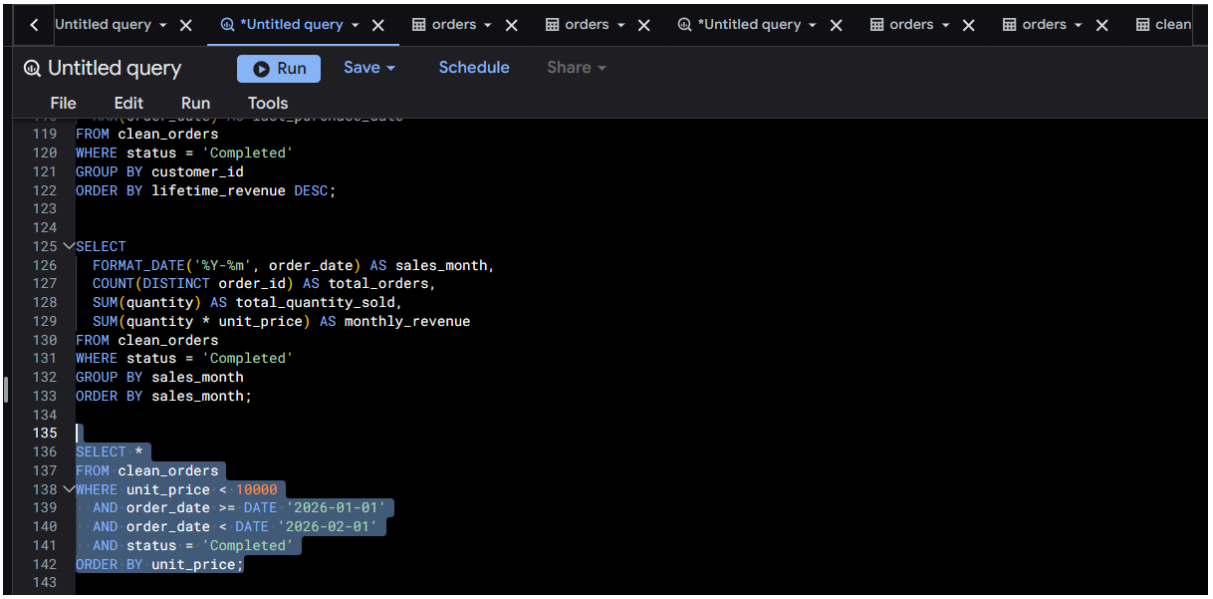
Query results

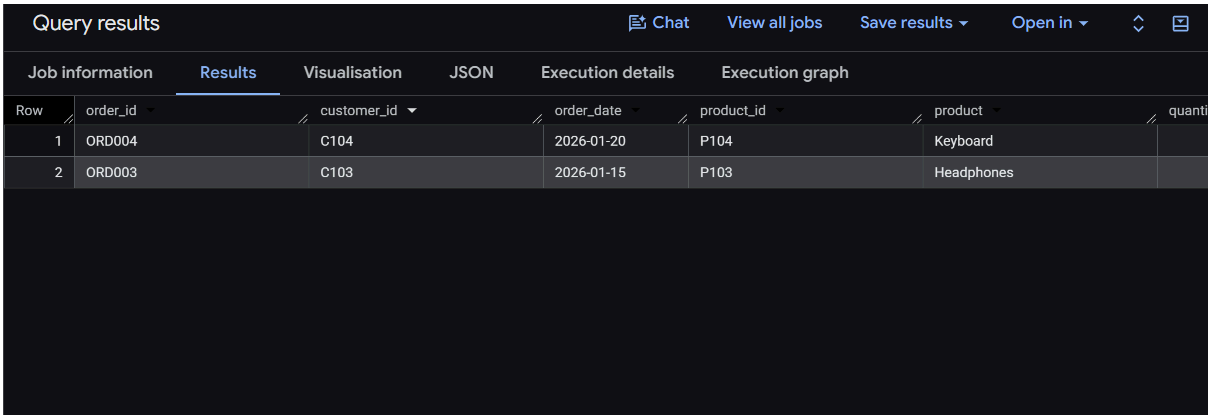
Chat | View all jobs | Save results | Open in

Job information | Results | Visualisation | JSON | Execution details | Execution graph

Row	sales_month	total_orders	total_quantity_sold	monthly_revenue
1	2026-01	5	9	156000
2	2026-02	5	10	137500
3	2026-03	4	9	114000
4	2026-04	5	11	152000

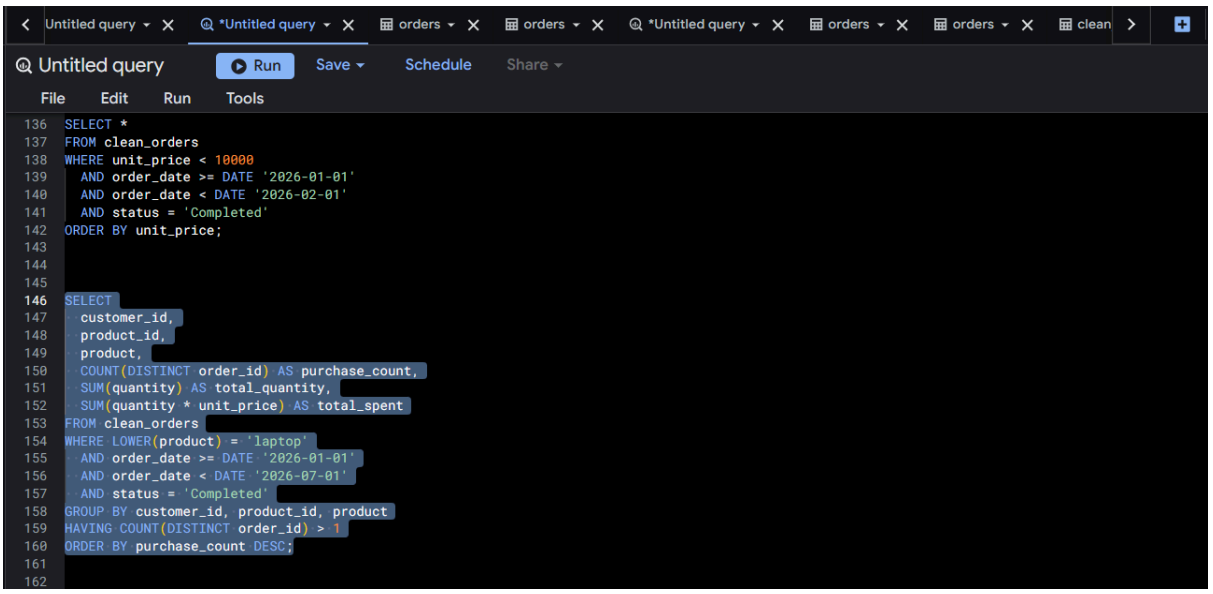
Task 8: find the unit price below 10000 in a particular month

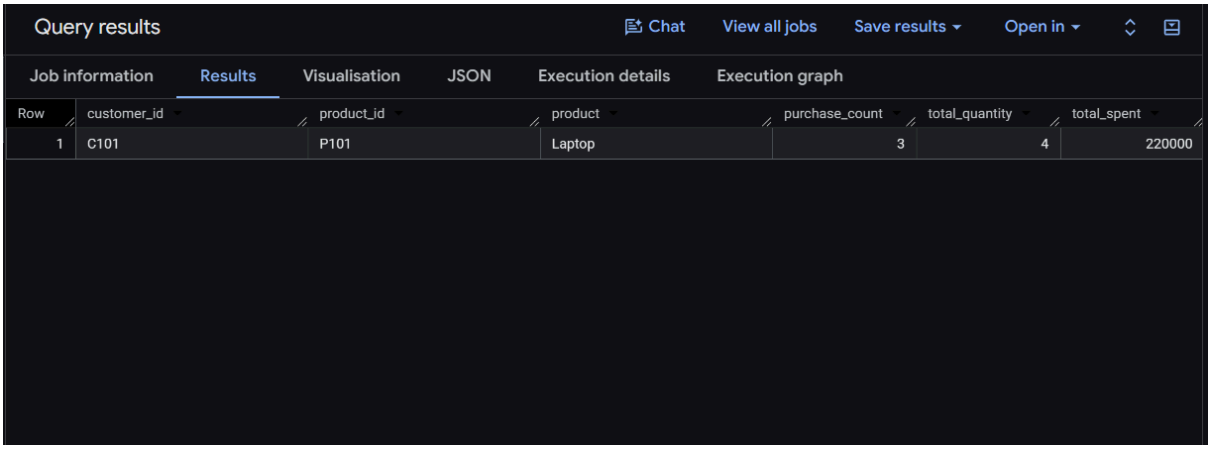




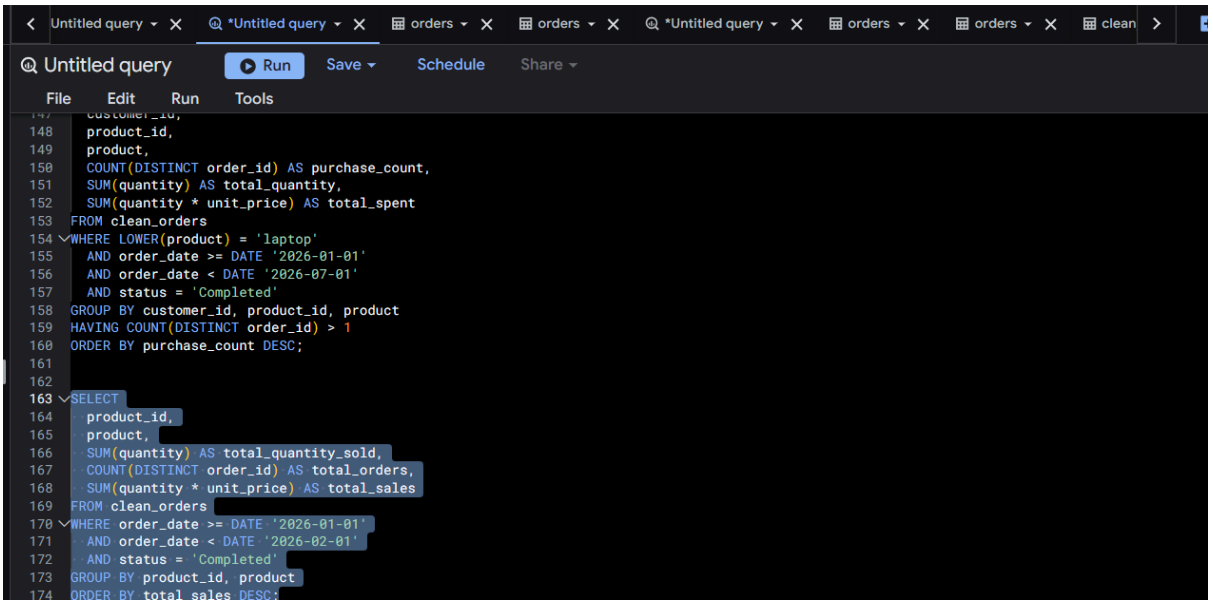
Row	order_id	customer_id	order_date	product_id	product	quanti
1	ORD004	C104	2026-01-20	P104	Keyboard	
2	ORD003	C103	2026-01-15	P103	Headphones	

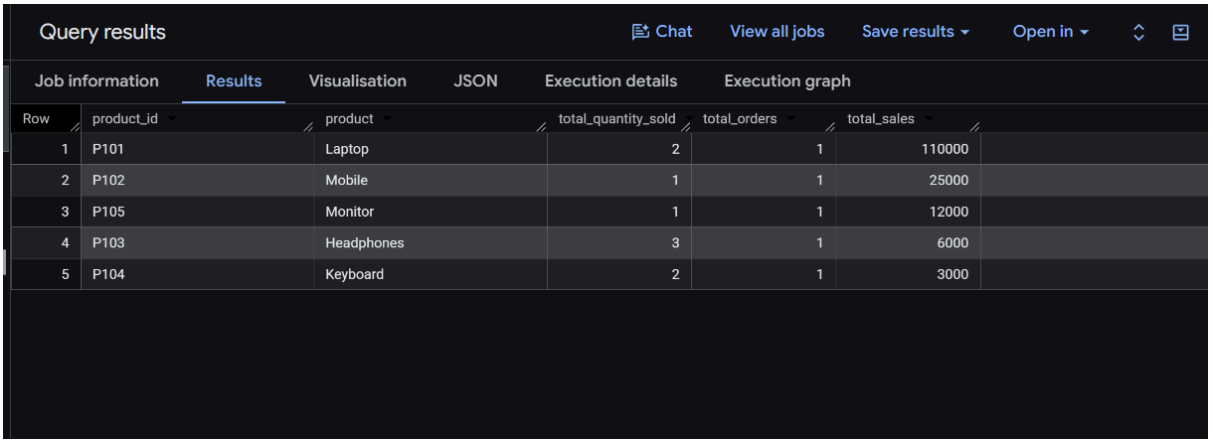
Task 9: find the customer who is purchased particular product multiple times in 6 months





Task 10: find the total sales report for each product in a particular month





Row	product_id	product	total_quantity_sold	total_orders	total_sales
1	P101	Laptop	2	1	110000
2	P102	Mobile	1	1	25000
3	P105	Monitor	1	1	12000
4	P103	Headphones	3	1	6000
5	P104	Keyboard	2	1	3000